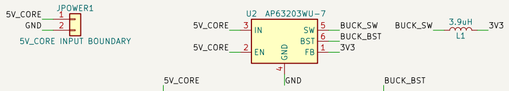
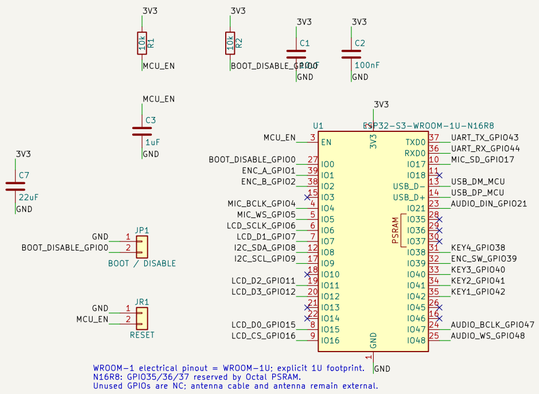
3V3 CONTROL RAIL: AP63203 fixed-output reference topology



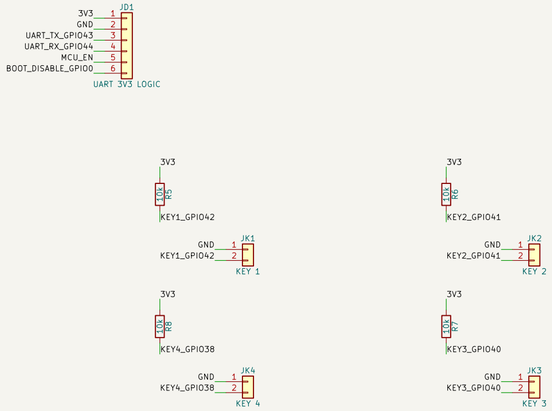
5V_CORE must be regulated / current-limited upstream.
Not raw USB-C / PD. 2A converter rating is not a USB power budget.
Do not parallel this 3V3 output with future Hub regulator.
JPOWER1 and L1 exact physical parts remain selection gates.

MCU + reset / BOOT / debug: existing firmware GPIO map



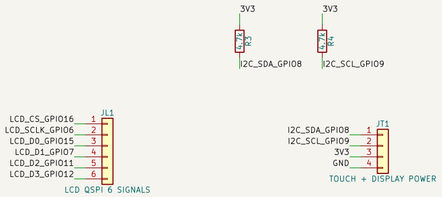
WROOM-1 electrical pinout = WROOM-1U: explicit 1U footprint.
N16R8: GPIO35/36/37 reserved by Octal PSRAM.
Unused GPIOs are NC; antenna cable and antenna remain external.

INPUTS: four keys + EC11, active-low to GND



No duplicate screen-press GPIO39.
Future independent screen press is not connected.
Connector A/B = running firmware GPIO1/GPIO2;
historical physical wire labels need assembly check.

DISPLAY / TOUCH HARNESSES



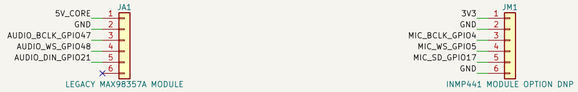
Harness pin numbers are new PCB pad numbers, not the LCD FPC order.
Preserve actual module silk CS/SCK/D0..D3 and SDA/SCL/3V3/GND.
One upstream I2C pull-up set only. QSPI return uses adjacent power harness.

MCU USB FULL-SPEED / future Hub downstream P6



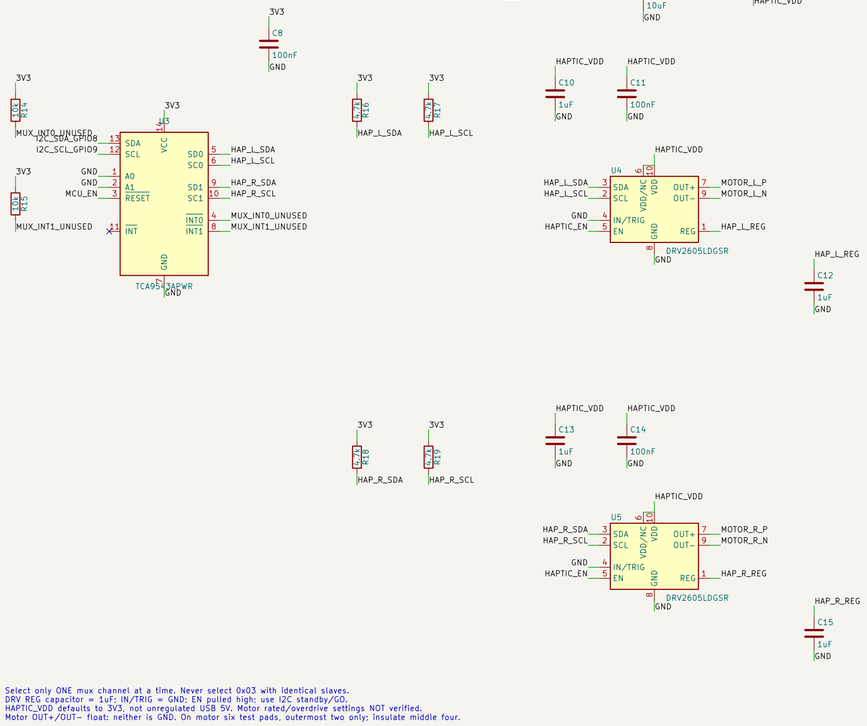
JU1 pin1 NC: no VBUS back-feed. Supply Core through JPOWER1.
This is an internal USB2 branch, not a USB-C receptacle / USB3 port.
External test adapter must provide connector-side ESD protection.
Keep D+/D- paired and short; final high-speed layout is a separate gate.

OPTIONAL AUDIO / MICROPHONE: keep modules external



JA1 is the already-tested I2S amplifier interface, not the motor driver.
MAX outputs drive only the speaker differential pair; never ground OUT-.
JM1 pin6 straps INMP441 L/R low for LEFT; module itself provides decoupling.
Microphone code optional, hardware not verified; default DNP.

HAPTICS: mux 0x70; channel0 LEFT / channel1 RIGHT; each driver 0x5A



Select only ONE mux channel at a time. Never select 0x03 with identical slaves.
DRV REG capacitor = 1uF; IN/TRIG = GND; EN pulled high: use I2C standby/GO.
HAPTIC_VDD defaults to 3V3, not unregulated USB 5V. Motor rated/overdrive settings NOT verified.
Motor OUT+/OUT- float: neither is GND. On motor six test pads, outermost two only; insulate middle four.

CORE CONTROL REVA — ELECTRICAL STUDY, NOT RELEASED FOR FAB

Not included: USB7206C Hub, PC/Dock power mux/PD, A1/A2/FAST-C, magnetic face, Dock high-speed connector.
These need exact receptacle/connector pin maps, source budgets and impedance/stackup design.
Screen assembly and EC11 use brackets/cables; motors attach to shell, not PCB pads.
Do not synchronize this new control sheet into the old CN7/CN10 schematic or overwrite the main placement PCB.

Sheet: /
File: Core-Control-RevA.kicad_sch
Title:
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.7 | Id: 1/1